

Zero weak gradients and constancy

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Scope

The weak equation makes local mollifications constant, giving local almost-everywhere constancy of the original function. Overlap and countable gluing arguments give global almost-everywhere constancy on an open preconnected domain. The separate continuous case upgrades this to pointwise constancy.

10 direct Cards + 0 reused prerequisites = 10 unique Cards. This count is a selected Card closure, not a source-declaration count.

[Route reading PDF](#) · [Preserved source exploration](#)

Cards in this route

- Key result: [Global almost-everywhere constancy on a preconnected domain](#)
- Continuous case: [A continuous function with zero weak gradient is constant](#)

[Read this route with prerequisites](#)

Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

[0](#) [1](#) [2](#) [3](#) [4](#) [5](#)

Level 0 (4 Cards)

Level 0

Gluing one almost-everywhere constant over a countable cover

Combines local exceptional sets using an inequality between restricted measures.

`MathlibAnnex.aeConstantOn_of_countableCover`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [From local to global almost-everywhere constancy](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Divergence as the trace of a derivative

Defines divergence without choosing coordinates.

`MathlibAnnex.divergence`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Vanishing weak gradient tested by divergence](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Identifying constants on an open overlap

Uses positive measure to find a point where both almost-everywhere equalities hold.

`MathlibAnnex.aeConstants_eq_of_open_overlap`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [From local to global almost-everywhere constancy](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Differentiating a local mollification through its kernel

Writes a directional derivative without differentiating the locally integrable input.

`MathlibAnnex.WeakGradient.localMollification_fderiv_apply`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Zero weak gradient gives zero derivatives of local mollifications](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1 (2 Cards)

Level 1

Vanishing weak gradient tested by divergence

Expresses a weak equation through compactly supported continuously differentiable vector fields.

`MathlibAnnex.WeakDivergenceZero`

Immediate Card prerequisites: [Divergence as the trace of a derivative](#)

Used by in this scope: [Zero weak gradient gives zero derivatives of local mollifications](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

From local to global almost-everywhere constancy

Separates agreement of constants by connectedness from countable measure-theoretic gluing.

`MathlibAnnex.exists_aeConstantOn_of_local`

Immediate Card prerequisites: [Gluing one almost-everywhere constant over a countable cover](#) · [Identifying constants on an open overlap](#)

Used by in this scope: [Global almost-everywhere constancy on a preconnected domain](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2 (1 Card)

Level 2

Zero weak gradient gives zero derivatives of local mollifications

Substitutes a translated smooth kernel into the weak equation and tracks the reflection sign.

`MathlibAnnex.WeakGradient.localMollification_fderiv_apply_eq_zero`

Immediate Card prerequisites: [Vanishing weak gradient tested by divergence](#) · [Differentiating a local mollification through its kernel](#)

Used by in this scope: [Local almost-everywhere constancy from the weak equation](#)

[Open Declaration Card](#) [Read exact source](#)

Level 3 (1 Card)

Level 3

Local almost-everywhere constancy from the weak equation

Turns constant mollifications into one almost-everywhere constant by choosing a common convergence point.

`MathlibAnnex.WeakDivergenceZero.nonempty_localAECOnstantAt`

Immediate Card prerequisites: [Zero weak gradient gives zero derivatives of local mollifications](#)

Used by in this scope: [Global almost-everywhere constancy on a preconnected domain](#)

[Open Declaration Card](#) [Read exact source](#)

Level 4 (1 Card)

Level 4

Global almost-everywhere constancy on a preconnected domain

Combines local weak-gradient rigidity with topological and countable gluing.

`MathlibAnnex.WeakDivergenceZero.exists_aeConstantOn`

Immediate Card prerequisites: [Local almost-everywhere constancy from the weak equation](#) · [From local to global almost-everywhere constancy](#)

Used by in this scope: [A continuous function with zero weak gradient is constant](#)

[Open Declaration Card](#) [Read exact source](#)

Level 5 (1 Card)

Level 5

A continuous function with zero weak gradient is constant

Removes the exceptional set using continuity and positivity of open sets.

`MathlibAnnex.WeakDivergenceZero.exists_eqOn`

Immediate Card prerequisites: [Global almost-everywhere constancy on a preconnected domain](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)